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On the Mechanism for Improved Passivation by Additions of Tungsten to Austenitic Stainless Steels*

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Abstract

In order to elucidate the role of W in improving the passivity of austenitic stainless steels, anodic polarization curves and X-ray photoelectron spectra have been obtained for pure W and W-containing steels passivated in 1 N HCl, 0.1 M NaCl, and in Cl^--containing H_3PO_4. Similar experiments were also performed on Cr-Ni steels passivated in WO_4^{2-} solutions. The results showed that pure tungsten by itself does not develop a protective layer and that alloying tungsten inhibits pitting corrosion, as do tungstate ions dissolved in neutral chloride solutions. In the passive layer, tungstate ions were found to be adsorbed without reduction, and in the case of alloying tungsten, this element is present under the +VI oxidation state. Alloying tungsten is thought to pass directly from the metal into the passive film, by interaction with water and formation of insoluble WO_3 which enhances the stability of oxide layers.

Introduction

It is well known that additions of Mo to austenitic stainless steels improve their passivity, particularly their resistance to pitting in chloride solutions.[1,2,12] It has also been shown that alloying W increases the corrosion resistance of austenitic 20Cr-25Ni steels in acid chloride solutions.[3,4] The primary purpose of W additions to stainless steels was to improve their mechanical resistance to abrasion corrosion in the wet process of phosphoric acid production.[3] While many attempts were made to clarify the role of molybdenum additions by XPS and AES surface analysis,[2,5-9] the mechanism for improved passivation by alloying tungsten has not been investigated, as far as we know in our survey of literature. The present study is an attempt to elucidate the role of W in the improvement of passive film stability. Previously, experimental 16Cr-14Ni stainless steels with varying amounts of W, up to 12 Wt%, were prepared, and polarization measurements in acid chloride and fluoride solutions were performed.[11] The results showed that in all cases the additions of tungsten to stainless steels reduce the anodic dissolution rate and enhance their passivation. As W and Mo are in the same column of the periodic table and have similar chemical properties, the question is whether alloyed W will have the same role as Mo in stainless steel. If so, the following propositions, as stated in Reference (2), are possible in interpreting the effect of W on passivity: (1) thickening of the passive films of the steels, (2) forming a heteropoly-acid film which is less hydrolyzed and more readily polymerized, (3) impeding the adsorption of Cl^- onto the surface of the steel by increasing the oxygen affinity of the steel, and (4) forming an amorphous oxide film with a glassy structure.

In the present investigation, it is proposed to answer the following questions:

1. Does W by itself form a protective film, without interaction with other elements such as Cr, Ni, Fe?
2. Does dissolution of alloyed W result in the formation of $WO_4^{=}$ ions which will inhibit the pitting process in neutral chloride solution? How effective is WO_4^{2-} as pitting inhibitor?

For these purposes, polarization measurements of pure W were carried out in 1 N HCl and in 40 Wt% H_3PO_4 containing 1000 ppm Cl^-. Then similar experiments were performed to compare the effect of W either as $WO_4^{=}$ ions dissolved in neutral chloride solution or as alloying element in the steels.

ESCA measurements were subsequently conducted on the passive films formed on the surface of W-containing Cr-Ni steels and on Cr-Ni steels passivated in $WO_4^{=}$ solutions. The experimental results were used to discuss the role of W on the basis of the above questions.

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TABLE 1 — Alloy Compositions

Alloy	Heat Treatment	C	Mn	Si	Ni	Cr	W
		Composition Wt%					
16-14	1150 C, 30 mn + W.Q. (water quenching)	0.006	0.43	0.33	14.40	16.30	0
16-14-2	1150 C, 30 mn + W.Q.	0.005	0.33	0.40	14.00	16.30	2.49
16-14-4	1150 C, 30 mn + W.Q.	0.006	0.30	0.41	14.10	16.50	4.81
16-14-8	1150 C, 30 mn + W.Q.	0.006	0.32	0.40	14.10	16.17	8.66
16-14-12	1150 C, 30 mn + W.Q.	0.006	0.30	0.43	14	16.22	12.40

Experimental Procedures

The five experimental austenitic 16Cr-14Ni alloys containing 0 to 12 Wt% were prepared in a vacuum furnace. Specimens were cut from rolled sheet of these alloys, having a thickness of 5 mm and a diameter of 12 mm. Their chemical compositions are given in Table 1. Solution treatment was performed by heating in nitrogen at 1150 C for 30 minutes followed by water quenching. For electrochemical measurement, the surfaces studied were mechanically polished with emery paper under water up to grade 1000, then were degreased in absolute ethyl alcohol, washed in deionized water, and dried by warm air blowing.

Polarization curves were obtained potentiodynamically at a scan rate of 20 mV/min, using a Tacussel electrochemical set composed of a potentiostat, a current recorder, and a high impedance electronic millivoltmeter.

The valency of alloyed W in the passive film formed on stainless steels in chloride solutions was investigated by means of ESCA spectroscopy, a surface sensitive technique which gives photoelectron spectra of elements in different valence states, in particular in metallic and oxide forms. The instrument was a Vacuum Generators ESCA 4. The samples were irradiated with Al Kα radiation, with a mean energy of 1486.6 eV. Photoelectron spectra of 4 d3/2 and 4 d5/2 levels of W were recorded. The binding energies at the peaks observed in the photoelectron spectra were corrected using the value of the energy of carbon 1s level (285 eV) observed in the background spectrum.

Results and Discussion

Electrochemical Behavior of Pure W

Anodic polarization curve of pure W in 1 N HCl and in 40 Wt% H_3PO_4 containing 1000 ppm Cl^-, at 60 C, are shown in Figure 1. It can be seen that corrosion potential of W is −0.10 V vs SCE, a value much higher than that of 16Cr-14Ni (−0.37 V), in 1 N HCl. Pure Mo, on the other hand, also has a high corrosion potential, −0.03 V in 1 N HCl at 25 C, as measured by other authors.[2] The main feature, however, is that W, as well as Mo, dissolves at a high rate in the passive region of the W containing 16Cr-14Ni steels or 20Cr-25Ni-5Mo steels. In other words, Mo and W alloying elements increase the pitting resistance of stainless steels in the potential region where they are highly active in pure form. This suggests that W does not protect by making a stable oxide layer, but only by interaction with other oxides. Sugimoto and Sawada[2] have shown that combination of Mo and Ni, as well as those of Mo and Fe, do not result in an increase in pitting resistance. It is thought that W improves the pitting inhibition, just like Mo, when the amount of Cr is above an adequate level in the alloy. It has

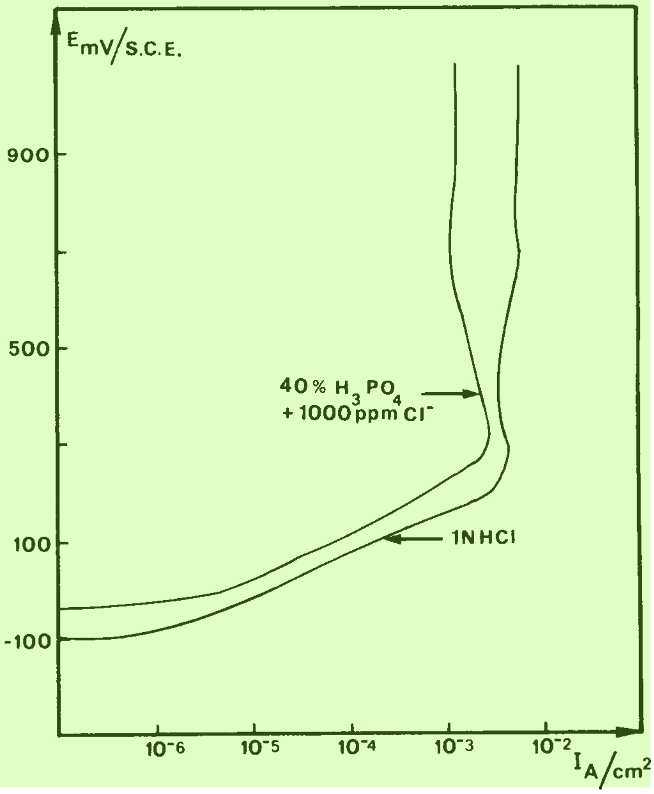

FIGURE 1 — Anodic behavior of pure tungsten.

been shown previously that alloyed W improves the pitting resistance of 20Cr-25Ni-4Mo steels.[3] This means that W might have an additive action with Mo and might have the same mechanism as Mo in forming a passive oxide layer with chromium.

Inhibitive Properties of Tungstate Ions

In the hypothesis that the dissolution of alloyed W results in the formation of WO_4^{2-} ions, polarization studies were carried out with 16Cr-14Ni steel in 0.1 M NaCl containing Na_2WO_4 at various concentrations, in order to evaluate the inhibitive effect of WO_4^{2-} in solution. Figure 2 shows the anodic polarization curves obtained in 0.1 M NaCl containing 0 to 0.025 to 0.1 M $WO_4^{=}$ at 60 C.

It can be seen that an increase in tungstate concentration from 0 to 0.1 M results in an elevation of corrosion potential from −360 to −230 mV (vs SCE) and at the same time a great decrease in passive current density.

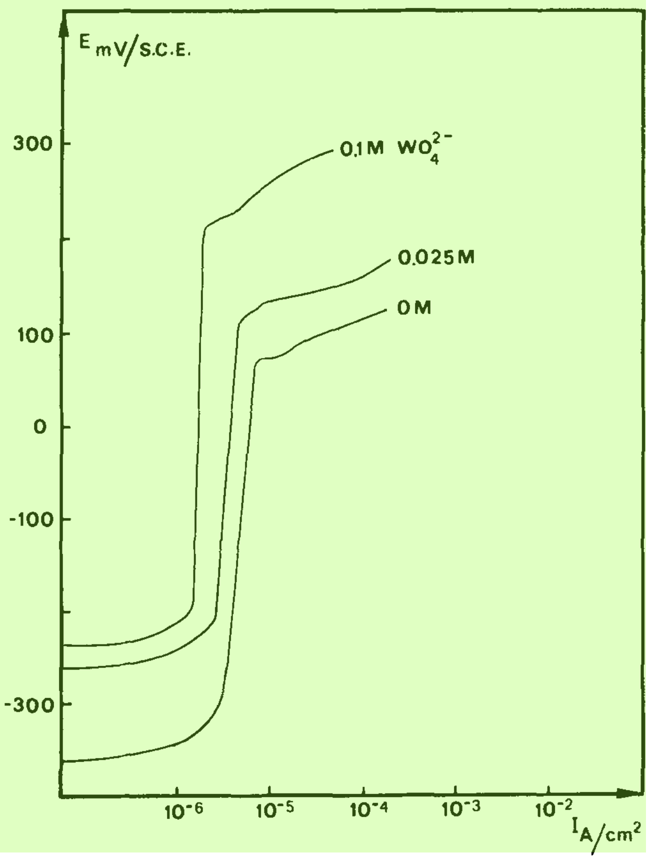

FIGURE 2 — Anodic behavior of 16Cr-14Ni steels in 0.1 M NaCl with various WO_4^{2-} additions.

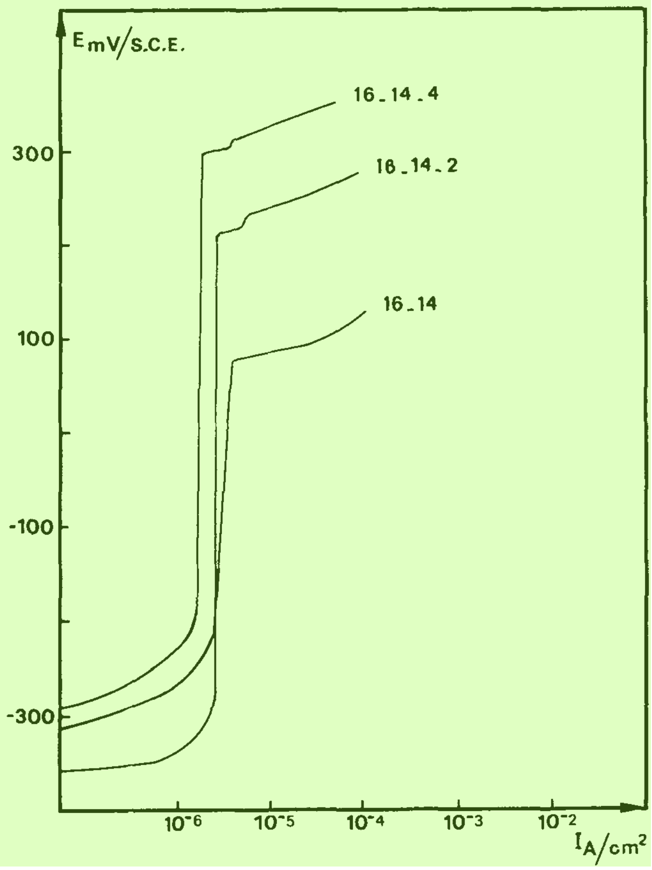

FIGURE 3 — Anodic polarization curves of W-containing steels in 0.1 M NaCl (curve numbering refers to Table 1).

The improvement of the passivity characteristics is further illustrated by the shifts of breakdown potentials to higher values: 80, 150, and 220 mV vs SCE, respectively.

For comparison, the effect of alloyed W was studied by determining the electrochemical behavior of W-containing steels in 0.1 M NaCl without tungstate. In Figure 3, it is shown that corrosion potential increases and passive current density decreases with increasing W content in the steels, but the effect is less pronounced than that produced by tungstate in solution. However, the effect of alloyed W in increasing the pitting potentials is greater.

These potential values are now 80, 210, and 300 mV vs SCE, respectively, for 0, 2, and 4% W in 16Cr-14Ni steels.

These results point out that the effect of alloying W is somewhat similar to adding WO_4^{2-} in 0.1 M NaCl solutions. There is, however, a little difference; while tungstate in solution has the effect of shifting the passivity range (*i.e.*, the potential difference between pitting potential and corrosion potential) in the noble direction of potential without changing it, alloying W has a greater tendency to extend this passivity range.

The action of $WO_4^{=}$ in solution may be explained by an adsorption process which has already occurred at the corrosion potential. According to Rosenfeld,[12] the oxide film formed through a natural process on the metal surface changes its properties when insignificant amounts of inhibitor are adsorbed. This modification is of structural and electrophysical nature rather than a variation of thickness or continuity of protective films. This proposed mechanism may be extended to the role of alloyed W. This element could be present in the oxide film which is instantaneously formed in air and could make a significant contribution to modify the passive properties of the film, by accepting oxide electrons and strengthening the oxide film crystal lattice.[12] At the corrosion potential, where the dissolution rate of W is very low, it is unlikely that alloyed W produces $WO_4^{=}$ in solution which later on will act as inhibitor by adsorption. It has been shown that during $CrO_4^{=}$ adsorption, chromium is reduced from (+VI to the +III valency state[12]) and that [Mo^{VI} may be reduced to Mo^{IV}.[17]] Therefore, ESCA studies were undertaken to determine what W oxidation states were present in the oxide film and whether there is a reduction of $WO_4^{=}$ during adsorption.

ESCA Study of the Passive Film

Prior to the examination, the passive film was developed on the surface by maintaining the sample for 15 minutes at −50 mV/SCE, a potential situated in the middle of the passive range, as can be seen in Figure 2.

For 16Cr-14Ni steel passivated in 0.1 M NaCl + 0.05 M $WO_4^{=}$ solution, ESCA spectrum of the surface layer (Figure 4) shows two peaks corresponding to tungsten, situated at 247.6 and 260.3 eV.

The presence of tungsten is attributed to adsorbed tungstate from the solution, but it is not known if this adsorption is accompanied by a reduction of tungstate to a lower oxidation state. In order to clarify this point, spectra of pure tungsten and pure tungstate were then obtained (Figure 5) by putting these substances together in the sample holder of ESCA apparatus. It can be seen that pure W is characterized by two peaks at 244.5 and 256.8 eV, while $WO_4^{=}$ or W (VI) have these two peaks shifted to 248.05 and 260.8 eV.

From this result, it can be stated that $WO_4^{=}$ inhibitor is adsorbed in the passive film under the oxidation state +VI, without electrochemical reduction.

Then the passive film developed on 16Cr-14Ni-4W steel in 0.1 M NaCl was analyzed. The spectrum obtained (Figure 6) shows that the two peaks related to tungsten in the film have the corresponding energies of 248 and 261 eV. It can be deduced that tungsten from solid solution in the stainless steel is present in the passive film under the +VI valent state.

In conclusion, the improvement of the corrosion resistance of 16Cr-14Ni steel by tungsten results from the presence of W (VI) in the passive layer, either by the adsorption

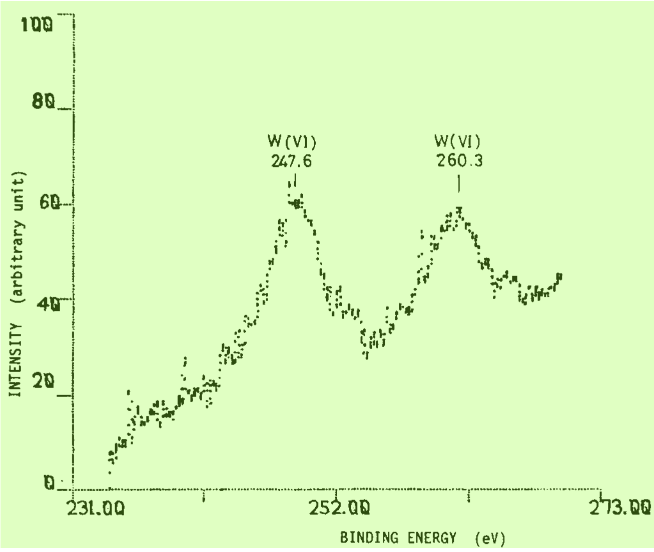

FIGURE 4 — ESCA spectrum of 16Cr-14Ni steel passivated in 0.1 M NaCl + 0.05 M WO_4^{2-}.

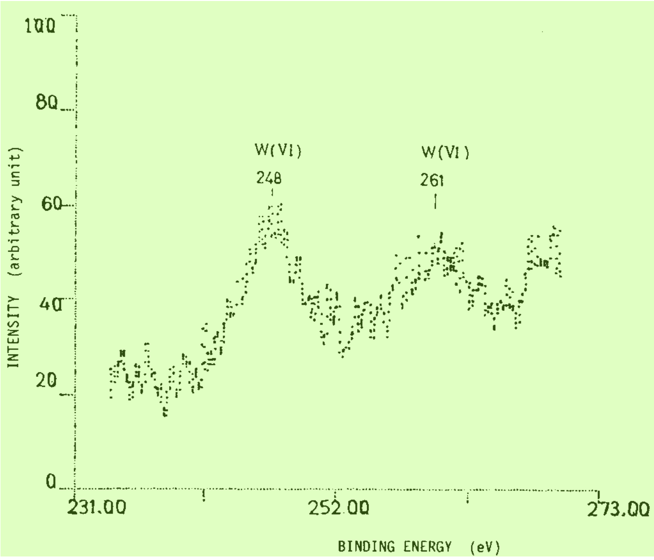

FIGURE 6 — ESCA spectrum of 16Cr-14Ni-4W steel passivated in 0.1 M NaCl.

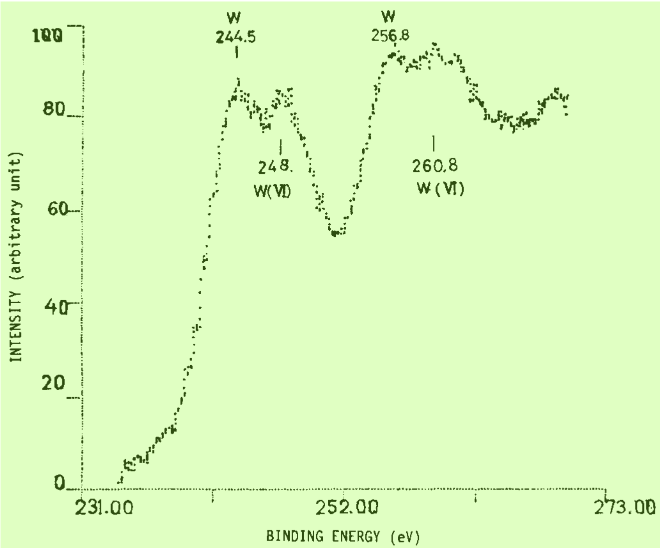

FIGURE 5 — Combined ESCA spectra of pure W and $NaWO_4$.

of dissolved tungstate in solution or by oxidation of metallic tungsten in the alloy. The mechanism by which W enters the passive film is not well explained. No experimental evidence has shown for W-containing alloys whether it is a direct solid-state reaction that takes place at the surface film or whether there is a previous step of W dissolution leading to tungstate formation which later will adsorb into the passive film. Some authors favor the latter process.[13] However, this formation of soluble $WO_4^{=}$ as an intermediate step cannot account for the role of W in the corrosion of stainless steels in acid chloride solution because the precipitation of $WO_4^{=}$ as sparingly soluble tungstic acid would be expected under these conditions. In confirmation of this, we observed yellow precipitates when adding $WO_4^{=}$ in 0.1 M HCl solution and these were subsequently identified as sparingly soluble tungstic acid H_2WO_4. With this inactive precipitate, suppression of the pitting inhibition was noted. On the other hand, alloyed tungsten has improved the resistance of 16Cr-14Ni steel already at the corrosion potential, as can be seen in Figures 7a and b. The higher the content of alloyed W, the more positive is the corresponding corrosion potential. This improvement of corrosion resistance could be brought about by W (VI) oxide present in the passive film. Furthermore, it is unlikely that W will dissolve as $WO_4^{=}$ in acid solution at potentials close to the corrosion potentials, but instead the following oxide formation is favored:

$$W + 3\,H_2O \rightleftharpoons WO_3 + 6\,H^+ + 6\,e^- \qquad (1)$$

The standard potential of this reaction is −0.09 V (NHE) or −331 mV (SCE),[14] while the corrosion potential of 16Cr-14Ni-12W is −330 mV (SCE) in 0.1 M HCl (Figure 7a). The effectiveness of WO_3 presumably present in the passive film can be interpreted in terms of its particular stability, and WO_3 occurs in nature as a component of many tungsten ores.[16] Furthermore, WO_3 is insoluble in water and the only acid that dissolves it is hydrofluoric acid. On the contrary, WO_3 is soluble in alkali hydroxide and carbonate solutions, yielding tungstate. Thus, the selective dissolution of WO_3 into tungstate in acid solutions, as an initial step of inhibition, seems improbable. Further, Pourbaix diagrams offer a thermodynamic confirmation of the acid stability of WO_3. A superposition of Pourbaix diagrams for Fe, Cr, Mo, and W (Figure 8) shows that WO_3 is the only insoluble product stable in the acid region. However, in this region, stainless steels without alloyed tungsten have been shown to be passive in acid solutions and surface analysis of the passive film reveals oxides or oxy-hydroxides containing Cr(III), Fe(III), Mo(VI),[7,10,18] or Mo(IV).[13] Therefore, thermodynamics for the individual elements seem less valid for alloys containing these elements, and stability regions of insoluble oxides may become larger when there are interactions with other oxides. This argument has been used by many authors[5,10,12] to interpret the beneficial effects of Cr and Mo. One possible mechanism is that these elements improve the quality of the bonding at the metal-oxide interface. Thus, a more efficient barrier layer is established, impeding the ionic or electronic transfer across it. This view can be extended to rationalize the effect of W as its physicochemical properties are very close to those of Mo; that is, the improved pitting corrosion resistance of stainless steels in NaCl solutions by W addition to the alloy can be explained by the induced insolubility of WO_3 in the neutral region, resulting from interaction with Cr and Fe oxides. In our ESCA study, Cr and Fe oxides were detected in the passive layer, but Ni oxide was not found, even after four-minute sputtering by ion bombardments. The binding energies of Fe 2p1/2 and 3p3/2 electron levels, as defined in ESCA spectra (not represented here), indicate the possible presence of α-FeOOH, according to the measurements reported elsewhere.[19] Cr 2p1/2 and 2p3/2 electron levels were found to correspond to those of Cr_2O_3, and the O 1s level may come from an M-OH bond (M denotes Fe or Cr).

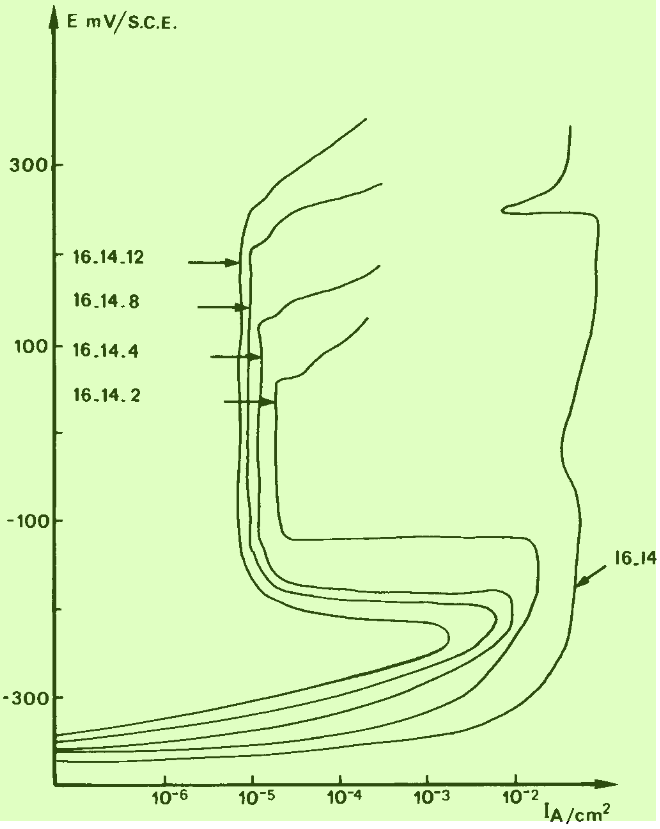

FIGURE 7a — Anodic polarization of W-containing steel in 0.1 M HCl.

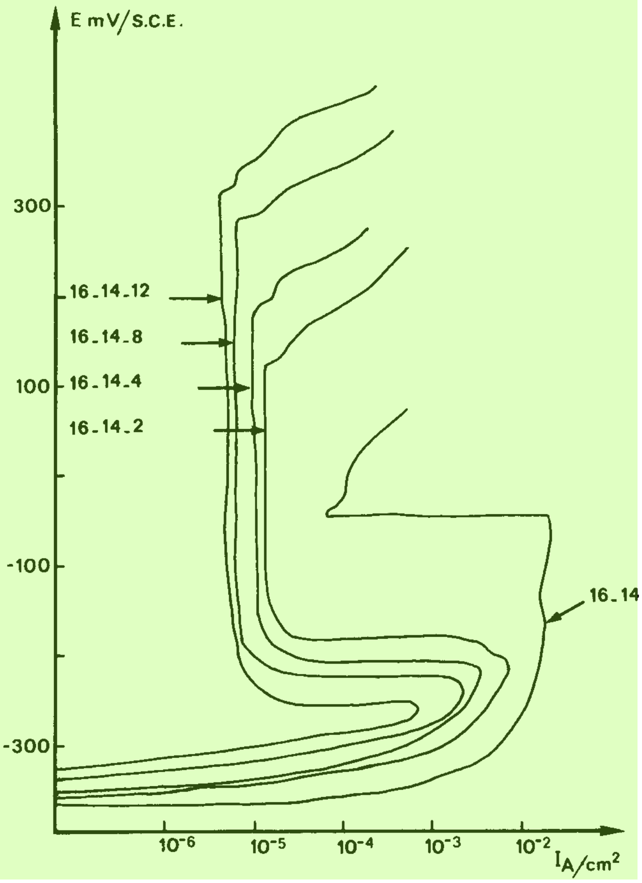

FIGURE 7b — Anodic polarization curves of W-containing steels in 0.2 M HCl.

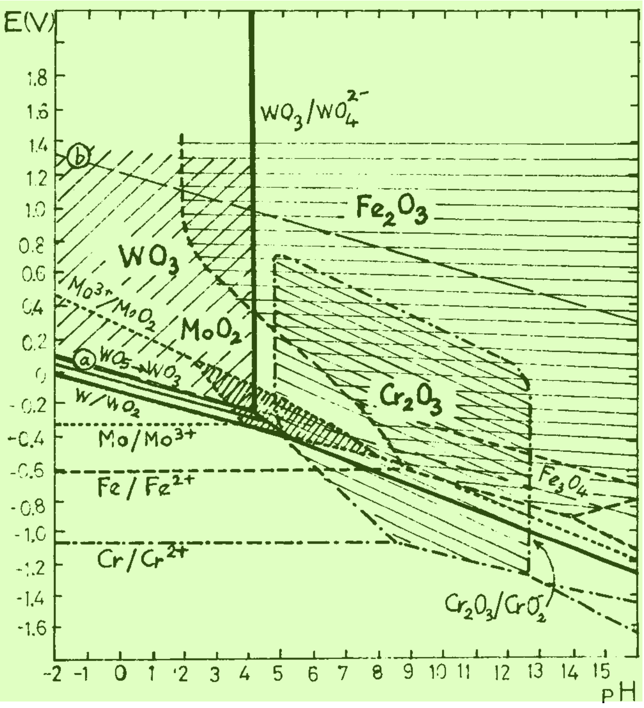

FIGURE 8 — Passivity regions of Cr, Fe, Mo, and W in Pourbaix potential-pH diagrams.

On the basis of these results, it can be suggested that WO_3, α-FeOOH and Cr_2O_3 are the main oxides which form a complex passive film having a very protective properties; consequently, the uptake of Cl^- from NaCl solution into the passive film was found to be very small. Actually, a pronounced ESCA spectrum of Cl 2p1/2-2p3/2 electron levels was obtained only after two hundred sweeps. These Cl binding energies in the film were confirmed as Cl^- since they were equal to those obtained from NaCl powder.

Conclusions

1. Pure tungsten does not develop a protective layer but, in alloy form, improves the pitting resistance of stainless steels in acid or neutral chloride solutions.

2. The tungstate ion dissolved in neutral chloride solution inhibits pitting corrosion of 16Cr-14Ni steel, as does alloying tungsten.

3. ESCA study of the passive film indicates that tungstate was adsorbed without modification of oxidation state and that alloying tungsten is also present in the passive layer in the + VI state.

4. In acid chloride solutions, W probably passes directly from the metal into the passive film, by interaction with water and formation of insoluble WO_3, rather than through a dissolution then adsorption process.

5. In neutral chloride solution, the beneficial effect of W is interpreted by the interaction of WO_3 with other oxides, resulting in enhanced stability and enhanced bonding of the oxide layers to the base metal.

Acknowledgment

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Inhibition of Carbon Steel Corrosion in Chemical Cleaning Solutions Containing Solid Magnetites*

*R. ROBERGE and R. GILBERT**

Abstract

A solvent composed of 8% EDTA, 2% citric acid, and 0.5% hydrazine, used at pH 6, and which was developed for the chemical cleaning of nuclear steam generators, was studied at 95 C for its agressiveness against carbon steel. Corrosion protection is improved by adding sulfur-free organic inhibitors to the pure solvent. In spite of this, when simulating the presence of corrosion products by adding magnetite, the effect of such inhibitors, as well as the corrosivity of the pure solvent, may be radically affected. Furthermore, different supplies of magnetite, whether they are added to the pure or the inhibited solvent, may have different solubilities and can drastically affect the corrosion rates. This might be explained in terms of the hydrazine evolution and/or the effective hydrazine content in solution during the test. It is found, nevertheless, that synergistic effects between surfactants and inhibitors could inhibit carbon steel corrosion, regardless of the supply of magnetite used. A mixture of benzotriazole, Triton X-100, and Chevron NI-W has been found useful as an inhibitor to chemical cleaning solvents of the type considered.

Introduction

In the last few years, a considerable amount of research on the chemical cleaning of nuclear steam generators has been performed in many laboratories. Among other aspects studied, the corrosion mechanisms involved in the use of the cleaning solvents,[1-5] and the development of process applications[6-12] have been extensively explored. The first successful chemical cleaning of an operational pressurized-water reactor type boiler was carried out at the end of 1979,[13] and the technique is now a recognized means of decelerating or stopping various corrosion phenomena detrimental to many units.

Applied to dissolve the corrosion products transported to or formed in a steam generator, solvents based on chelating agents are aggressive toward unalloyed steels and corrosion inhibitors are therefore added to protect them. The best inhibitors used so far for that purpose are often based on sulfur compounds, but the possibility that these might be detrimental to Inconel 600, which proves susceptible to intergranular stress corrosion cracking in the presence of some sulfur-containing species,[14,15] has created the need to look at new substitutes. This aspect of materials protection through the addition of organic corrosion inhibitors to chemical cleaning solvents based on chelating agents is still grossly underdeveloped. Hausler[16] studied the corrosion mechanism on steel of a solvent based on 10% EDTA and 1% hydrazine. From a parametric study, he interpreted the general inhibition mechanism observed with and without magnetite and a commercial inhibitor, and found that the corrosion rates in such environments were controlled by some competitive complexation reactions involved in the interphase and controlled by the following local conditions: pH, EDTA, and iron concentrations.

It is obvious, therefore, that each new chemical cleaning solvent based on even slightly different chemicals or different concentrations of the same chemicals must be examined for its corrosiveness. These differences may be thought *a priori* to modify the reaction kinetics in the interphase. Similarly, in the case of the dissolution of various types of oxide, it may also be expected that dissolution of magnetites formed *in situ* may result in corrosion rates different from those obtained in laboratory experiments performed on commercial or synthetic magnetites.

For this purpose, experiments were conducted on a number of sulfur-free organic corrosion inhibitors and surfactants in a solvent based on EDTA, citric acid, and hydrazine (hereafter referred to as NPD solvent), which was originally

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